

Path Smoothing and Turning-Maneuver Optimization for Differential-Drive Mobile Robots

Intelligent Control, Robotics and Automation — Autonomous Systems (AMR, AGV, UAV)

Abstract—Global planners in ROS 2/Nav2 often produce piecewise-linear paths with sharp heading changes that slow differential-drive robots or require in-place rotations. This paper proposes Path Smoothing and Turning-Maneuver Optimization (PSTMO), a post-processing method that conditions the input path, constructs quintic Bézier transitions from two geometry-based trimming distances, rejects candidates violating kinematic, timing, or swept-footprint constraints, and uses dynamic programming to select compatible transitions and rotations. Evaluation in seven Gazebo environments used NavFn A*, NavFn Dijkstra, Theta*, Smac 2D, and Smac Hybrid, with Raw, Simple, Savitzky–Golay, and Constrained as baselines. In 34 complete planning comparisons, PSTMO reduced mean path length, maximum curvature, and squared-curvature integral by 0.64%, 75.29%, and 81.44% relative to Simple. Closed-loop evaluation comprised 175 isolated runs. All five methods completed 34 of 35 planner–environment cases; the shared failure was Smac Hybrid in Warehouse Dispatch. Across successful paired cases, mean travel times were 57.120, 54.449, 56.677, 57.293, and 53.148 s for Raw, Simple, Savitzky–Golay, Constrained, and PSTMO. Thus, 170 runs reached the goal without collision-monitor intervention, while PSTMO achieved the lowest mean travel time and substantially reduced turning severity.

Keywords—differential-drive mobile robot; path smoothing; turning-maneuver optimization; Bézier transition; ROS 2/Nav2.